AP CALCULUS AB

Unit 1 • Lesson 1.9 notes

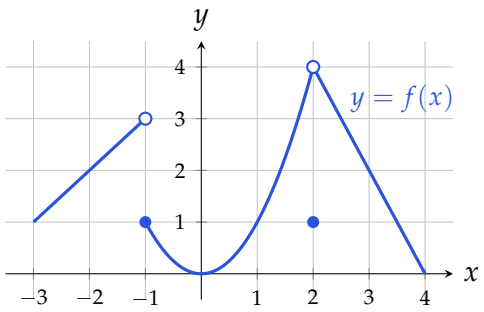
Lesson 1.9 • Connecting Multiple Representations of Limits

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Combine limits read from graphs, tables and formulas.
- Evaluate limits of composite functions, paying attention to the side of approach.

Vocabulary. representation, composite limit, inner function, outer function.



1. Combining representations

Evaluate each piece with the representation you are given, then combine with the limit properties.

Example 1

Using the graph of f above and $g(x) = x^2 - 1$, find $\lim_{x \to 2} \dfrac{f(x)}{g(x)}$.

Solution. $\lim_{x \to 2} f(x) = 4$ (graph), $\lim_{x \to 2} g(x) = 3$ (formula). Quotient: $\frac{4}{3}$.

2. Composite limits

Key idea

To find $\lim_{x \to c} f(g(x))$: first find $b = \lim_{x \to c} g(x)$ and note *from which side* $g(x)$ approaches b; then find the corresponding limit of f at b.

Example 2

Using the graph of f, find $\lim_{x \to 0} f(x - 1)$ and $\lim_{x \to 0^+} f(-1 - x)$.

Solution. As $x \to 0$, $x - 1 \to -1$ from both sides; f has different one-sided limits at -1, so $\lim_{x \to 0} f(x - 1)$ does not exist. As $x \to 0^+$, $-1 - x \to -1$ from the *left*, so $f(-1 - x) \to 3$.

Example 3 • Table and graph

A table shows $h(x) \to 2$ as $x \to 3$, with $h(x) > 2$ on both sides. Find $\lim_{x \to 3} f(h(x))$.

Solution. The inputs to f approach 2 from the right, and $\lim_{x \to 2^+} f(x) = 4$. So the limit is 4 (not $f(2) = 1$).

Common mistakes

- Evaluating f *at* the inner limit instead of finding the limit of f there.
- Ignoring the direction from which the inner function approaches.

Quick check

With the graph of f: 1. $\lim_{x\to 1}\left(f(x)+3x\right)$ 2. $\lim_{x\to -1^-}\left(2f(x)\right)$

Answers. 1. $1+3=4$. 2. 6.